

# Context-Aware Graph Neural Network for Skin Lesion Classification

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## Abstract

The accurate classification of skin lesions, particularly melanoma, is vital for the early detection and effective treatment of skin cancer. Although deep learning models such as convolutional neural networks (CNNs) have achieved remarkable success in dermoscopic image analysis, they often overlook valuable structured metadata (e.g., patient demographics, lesion location, and type) that provide essential diagnostic context. We present a *graph-driven multimodal framework* that jointly models visual and metadata information for skin lesion classification. Our approach uses a frozen CNN backbone to extract deep visual representations via dual pooling (mean and max), which are concatenated with encoded metadata and partitioned into subspaces. These subspaces are treated as nodes within a graph, where a graph neural network (GNN) captures intra-sample dependencies between feature subspaces and clinical attributes to refine lesion representations. Experiments on four public benchmarks: ISIC2024, HAM10000, PAD-UFES-20, and HIBA, demonstrating consistent performance gains over several state-of-the-art (SOTA) approaches, with relative accuracy improvements ranging from +0.5% to +8.8% in datasets. The results highlight the potential of graph-based modeling of metadata and image features to build more robust and clinically informed skin cancer classifiers.

## CCS Concepts

• Computing methodologies → Artificial intelligence;

## Keywords

Skin Lesion Classification, Clinical Metadata Integration, Subspace Modeling, Multimodal Fusion, Graph Neural Networks, Medical Image Analysis

## ACM Reference Format:

Muhammad Azeem, Saqib Nazir, Amr Ahmed, and Ardhendu Behera. 2026. Context-Aware Graph Neural Network for Skin Lesion Classification. In *The 41st ACM/SIGAPP Symposium on Applied Computing (SAC '26)*, March 23–27, 2026, Thessaloniki, Greece. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3748522.3779958>



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ACM ISBN 979-8-4007-2294-3/2026/03

<https://doi.org/10.1145/3748522.3779958>

## 1 Introduction

Skin cancer remains one of the most prevalent and life-threatening malignancies worldwide. Recent epidemiological studies report more than 330,000 new cases of melanoma and more than 55,000 deaths each year [1]. Therefore, an early and accurate diagnosis is crucial to improving patient outcomes and survival rates. In recent years, deep learning-driven computer-aided diagnosis (CAD) systems have shown remarkable success in automating the classification of skin lesions from dermoscopic and clinical images [2, 3]. However, most of these approaches rely exclusively on visual information and often overlook structured metadata, such as patient demographics, lesion location, and anatomical type, which can provide valuable clinical context and significantly improve diagnostic reliability [4, 5]. Despite these advances, two critical challenges remain unexplored. (1) *Metadata underutilization*: most existing frameworks focus on visual cues while neglecting rich clinical metadata that carry strong diagnostic priors. (2) *Ineffective fusion strategies*: when metadata are included, fusion typically occurs at a fixed stage (early or late) and fails to capture dynamic, fine-grained interactions between image and metadata representations [6, 7]. Overcoming these limitations is critical to developing clinically meaningful and generalizable diagnostic systems. Previous works have attempted to address these issues individually. Graph-based models [8, 9] capture spatial patterns within the image but typically ignore metadata, while attention-based fusion models [10] incorporate metadata but overlook relational structure, treating CNN embeddings as monolithic features. Graph neural networks (GNNs) have emerged as powerful alternatives capable of capturing nonlocal dependencies and relational structures of skin lesions [8]. Existing approaches either rely on fixed graph topologies that fail to generalize across datasets or lack mechanisms to handle heterogeneous metadata. Furthermore, metadata such as patient age, sex, lesion site, and anatomical context provide rich semantic cues that, if properly integrated, can substantially improve classification performance. For example, [11] shows strong accuracy in ISIC data but does not systematically encode the patient's context. In contrast, [10] merges metadata by attention (MetaBlock) but ignores spatial structure in CNN features, limiting cooperative reasoning in modalities. To bridge this gap, we propose a unified graph-driven architecture that jointly models intra-image spatial dependencies and inter-modal correlations between image features and metadata. Our design uses novel message passing and aggregation to

refine subspace features dynamically, enabling *context-aware* and *relational reasoning* across both *visual* and *clinical* dimensions.

**Motivation and Contribution:** Fine-tuning pre-trained CNNs remains a dominant strategy in skin lesion classification; this paradigm faces critical, often unsupported, limitations that directly motivate the research problem addressed in this work. Firstly, the method often suffers from a significant *domain mismatch* when adapting models trained on large natural image collections (like ImageNet) to the specific, high-resolution textures of medical dermoscopic images [12]. This discrepancy can lead to the disruption of rich, transferable representations and suboptimal performance in the target domain [1]. Secondly, given the typically small and class-imbalanced nature of many medical datasets, fine-tuning is prone to *overfitting* to minor distribution shifts, which severely hinders the system’s generalizability and clinical reproducibility [2, 3]. Finally, the process of back-propagating through massive, state-of-the-art backbones is *computationally inefficient*, placing a high burden on clinical deployment and experimentation, especially compared to more lightweight adaptation strategies [5]. Therefore, a key motivation of our work is to address the limitations of conventional fine-tuning approaches, which often exhibit reduced efficiency and limited generalization, as well as those of existing fusion strategies, which tend to underutilize metadata and rely on rigid integration mechanisms.

Our framework addresses these challenges by adapting the frozen backbone, combined with graph-based relational learning. Instead of fine-tuning, we extract visual embeddings from a frozen CNN trained on large natural image corpora and adapt them to the target medical domain via graph reasoning. Specifically, the CNN feature space is divided into  $N$  non-overlapping subspaces, each treated as a node in a fully connected graph. A graph neural network then models relationships and dependencies among these nodes, learning structured adaptations within the feature space without altering the backbone parameters.

To enrich contextual reasoning, *clinical metadata are integrated directly at the subspace level*, allowing each node to encode both visual and patient-specific attributes. This enables the model to capture fine-grained correlations between the extracted image features and metadata cues such as patient age, sex, or lesion location information. The resulting system is lightweight, computationally efficient, and more interpretable, providing a principled approach to achieving clinically robust lesion classification without the need for costly fine-tuning.

This graph-driven fusion strategy enables the propagation of clinically relevant information across the graph, enhancing the discriminability of the learnt representations and making them more context-aware. The overall pipeline of our approach is illustrated in Figure 1. We evaluated the proposed framework on four widely used skin lesion benchmarks: International Skin Imaging Collaboration (ISIC) 2024 [13], HAM10000 [14], PAD-UFES-20 [15], and HIBA [16]. We achieve consistent gains in classification accuracy and robustness across datasets and lesion types. Our main contributions are summarised as follows.

- **Graph-driven multimodal fusion:** We introduce a novel framework that integrates visual features and clinical metadata at the subspace level. By modeling interactions between

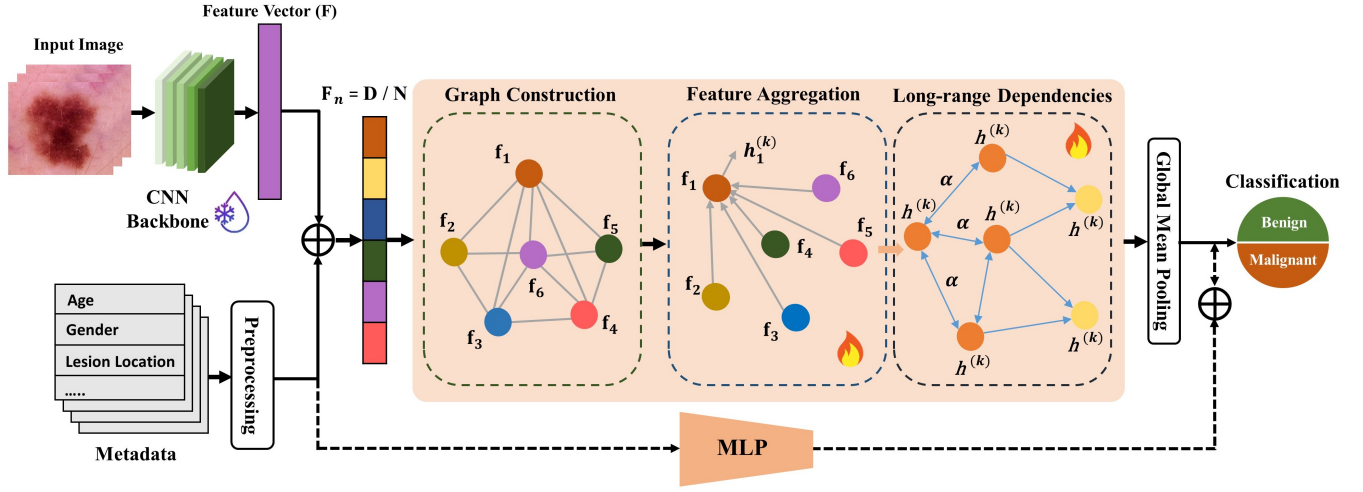
CNN-derived feature subspaces and patient-level attributes through a fully connected graph, our approach propagates clinically meaningful information, leading to more discriminative and context-aware lesion representations.

- **Unified feature-to-graph transformation:** We design a principled mechanism that jointly embeds heterogeneous metadata and visual subspaces into graph nodes via systematic subspace partitioning. This enables the GNN to capture both local dependencies within image features and long-range correlations across modalities.
- **Inductive GNN formulation for lesion classification:** We propose an inductive message-passing formulation that improves generalization across datasets and lesion categories without relying on transductive assumptions or fixed graph structures.
- **Comprehensive empirical validation:** Extensive experiments on four public benchmarks demonstrate consistent improvements over SOTA approaches, with average relative gains of +1.31% in accuracy, +1.06% in precision, +2.32% in recall, and +1.24% in F1-score (where possible), confirming the robustness and effectiveness of the proposed approach.

## 2 Related Work

Recent advances in deep learning have driven significant progress in automated skin cancer classification, with CNN serving as the dominant backbone. Architectures such as InceptionV3, VGG, EfficientNet, ResNet, and DenseNet have been widely adopted for their proven performance in image-based skin lesion classification tasks [17, 18]. For example, [19] introduced HierAttn, a lightweight network that employs hierarchical attention and deep supervision across multiple branches, achieving 96.7% precision on the ISIC 2019 dataset. Similarly, [20] combined CNNs with Vision Transformers (ViT) to use local and global representations, reporting 94.3% and 92.5% accuracy on HAM10000 and ISIC 2019, respectively. [21] further enhanced feature diversity by integrating discrete wavelet decomposition with an EfficientNet backbone, achieving 92.2% accuracy on ISIC 2017. More recently, [22] proposed MABSCNET, a CNN–Transformer fusion framework augmented by metaheuristic hyperparameter optimization and ensemble classification, yielding 94.74% accuracy on ISIC 2020. These studies collectively highlight the growing maturity of deep CNN and hybrid CNN–ViT models in capturing complex visual cues for the classification of lesions.

**Multimodal Fusion Approaches:** Integrating visual and non-visual cues, such as patient demographics or lesion metadata, has emerged as a promising direction to improve both accuracy and interpretability in skin cancer diagnosis. Multimodal fusion enables models to combine complementary sources of information that reflect clinical decision-making. Several approaches have explored the fusion of traditional and deep learning features from heterogeneous modalities. For example, [10] introduced the MetaBlock module, which adaptively incorporates metadata into CNN features to enhance salient visual representations, yielding improved results on ISIC 2019 and PAD-UFES-20. Likewise, [23] proposed skin lesion image metadata pre-training (SLIMP), a nested contrastive learning framework designed to jointly embed lesion images, lesion-level



**Figure 1: Overview of our framework.** The feature vector  $F \in \mathbb{R}^D$  from a frozen CNN backbone is concatenated with the preprocessed metadata before graph construction. It is then partitioned into  $N$  subspaces representing nodes (e.g.,  $N = 6$ ) of a fully-connected graph, which undergoes (i) relation-aware feature aggregation, and (ii) long-range dependencies modeling, where multi-hop message passing captures global dependencies. The final graph-level embedding is obtained using global mean pooling and passed to a classifier to predict lesion type (benign or malignant). As an ablation study (shown with dotted lines), metadata features are first passed through a multilayer perceptron (MLP), and the resulting feature vector is fused (late fusion) with the graph-processed frozen image feature before the final classification. It is discussed in the ablation studies section.

metadata, and patient-level clinical features. SLIMP consistently outperformed conventional image-only pretraining strategies across the HAM10000, PAD-UFES-20, and HIBA datasets. Although these fusion methods demonstrate the value of multimodal learning, their designs are typically limited to *shallow integration strategies* (either early, late, or fixed stage fusion), which fail to capture dynamic and structured relationships between visual and clinical modalities. Consequently, such approaches often underutilize metadata, leading to suboptimal generalization and limited clinical insight.

**Graph-based Learning:** These approaches have recently gained traction for skin lesion analysis, offering a principled way to model spatial, morphological, and contextual relationships between image regions or samples [24]. In these frameworks, the lesion regions are represented as nodes, and their spatial or semantic relationships are encoded as edges. For example, [11] employed a GraphSAGE-based CNN that achieved 90% accuracy on the ISIC Archive dataset, demonstrating the potential of GNN to capture non-local dependencies. Building on this idea, [25] introduced the Feature-Enhanced Gated Graph Neural Network (FEGGNN), which constructs inter-sample graphs based on feature similarity and incorporates asymmetric CNN encoders for feature extraction, achieving 84.9% accuracy on the Derm7pt dataset. Despite their effectiveness, most graph-based methods focus solely on *intra-image* or *inter-sample* relations and ignore the valuable clinical metadata that can guide relational reasoning. In contrast, existing multimodal fusion approaches incorporate metadata but lack structured graph reasoning.

**Summary and research gap:** While CNN and CNN-ViT hybrids excel at learning visual representations, they *neglect the rich contextual priors* present in metadata. In contrast, multimodal fusion methods often employ shallow or static fusion schemes and graph-based

models, though relationally powerful, typically exclude clinical information.

To bridge these gaps, we propose a *unified graph-driven framework* that performs an early and structured fusion of image features and metadata. Our method enables dynamic interactions across modalities via message passing, allowing clinically relevant information to propagate across feature subspaces. This approach improves both the discriminative power and the clinical interpretability of lesion classification, surpassing current state-of-the-art paradigms.

### 3 Methodology

Our novel graph-driven multimodal architecture for skin lesion classification is illustrated in Figure 1. The framework integrates image features extracted from a frozen CNN backbone with structured metadata, enabling context-aware reasoning through a graph neural network (GNN).

**Feature Extraction:** A pretrained CNN (e.g., ResNet or EfficientNet) is used as a frozen backbone to extract visual features from each dermoscopic image. From the final convolutional layer, we obtain two complementary representations: (1) the global average pooled feature map ( $F_1$ ), and (2) the global max pooled feature map ( $F_2$ ). Metadata consists of categorical and numerical attributes, including patient demographics (e.g., age, sex), lesion location, and lesion type. Categorical variables are one-hot encoded, while numerical variables are normalized using z-score standardization, resulting in a processed metadata vector  $F_3$ . The final representation of the feature is then formed by concatenating these components as  $F = [F_1, F_2, F_3] \in \mathbb{R}^D$ , where  $D$  denotes the overall dimensionality of the feature vector.

**Graph Construction:** Given a set of  $\mathcal{P}$  images  $I = \{I_p | p = 1, \dots, P\}$  with corresponding feature vectors  $F_p$  and class labels  $Y_p$ , the objective is to learn a mapping function  $\mathcal{F}$  that predicts the lesion labels  $\hat{Y}_p = \mathcal{F}(F_p)$ , minimizing the loss  $\mathcal{L}(Y_p, \hat{Y}_p)$ .

In this work,  $\mathcal{F}$  represents an end-to-end deep network that integrates a graph-driven learning mechanism to advance skin lesion classification. For clarity, we simply denote the feature vector of a given image as  $F$ . Unlike conventional CNN-based classifiers that process  $F$  as a flat vector, our model transforms it into a context-aware representation by explicitly modeling interactions between its feature subspaces. The network  $\mathcal{F}$  thus comprises two key components: (1) relation-aware subspace feature aggregation and (2) long-range dependencies modeling between subspaces. To achieve this, the feature vector  $F$  is uniformly divided into  $N$  contiguous subspaces,  $F = (f_1, f_2, \dots, f_N)$ , where each subspace  $f_n \in \mathbb{R}^{D/N}$  represents a distinct portion of the feature embedding. We then construct a fully connected graph  $G = (V, E)$ , where each node  $v_n \in V$  corresponds to a subspace  $f_n$ , and each edge  $e_{i,j} \in E$  represents a potential dependency between subspaces  $f_i$  and  $f_j$ . A GNN is applied to model these relationships, enabling message passing among nodes to yield enriched, context-aware subspace representations.

**Metadata Fusion:** To ensure that metadata are seamlessly integrated within the graph representation, we perform vector-level fusion before graph partitioning. Specifically, metadata features are concatenated with CNN features to form a joint representation:

$$F = [F_{\text{cnn}}, F_{\text{meta}}] \in \mathbb{R}^D, \quad D = D_{\text{cnn}} + D_{\text{meta}}. \quad (1)$$

This ensures that metadata are directly embedded within the same latent space as visual features rather than represented as separate nodes. The concatenated vector  $F$  is then divided contiguously into  $N$  sub-vectors  $(f_1, f_2, \dots, f_N)$ , forming the nodes of the graph. When  $D$  is not evenly divisible by  $N$ , zero-padding is applied to maintain equal node dimensionality.

**Relation-aware Feature Aggregation:** To enable effective propagation of information across feature subspaces, we employ an *inductive neighborhood-based graph aggregation* inspired by [26]. This mechanism allows each subspace node to update its representation based on both its own state and that of its neighbors, thereby capturing local relational dependencies within the feature graph.

Let node  $i$  correspond to a subspace feature vector  $f_i$ . At iteration  $k$ , its representation is denoted by  $h_i^{(k)}$ , while  $\mathcal{N}(i)$  denotes its neighborhood set. The base representation is initialized as  $h_i^{(0)} = f_i$ . The statistical aggregation function that updates the node  $i$  in  $k^{\text{th}}$  iteration by computing the transformed feature  $h_i^{(k)}$  by summarizing the information about the neighborhood and self-attention to itself. This is given by:

$$h_i^{(k)} = \text{ReLU} \left( \underbrace{W_{\text{self}} h_i^{(k-1)}}_{\text{self-subspace}} + \underbrace{W_{\text{neigh}} \cdot \text{AGG}(\{h_j^{(k-1)} : j \in \mathcal{N}(i)\})}_{\text{neighbour context}} \right) \quad (2)$$

where  $W_{\text{self}}$  and  $W_{\text{neigh}}$  are learnable transformation matrices applied to self- and neighbor representations, respectively. The operator  $\text{AGG}(\cdot)$  is a *permutation invariant aggregation function* that summarizes information from neighboring nodes, ensuring robustness

to graph topology permutations. The ReLU non-linearity promotes sparsity and stabilizes learning.

We explore two statistical aggregation functions, *mean* and *max*, formally defined as:

$$\text{AGG}_{\text{mean}}(\{h_j\}) = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j, \quad \text{AGG}_{\text{max}}(\{h_j\}) = \begin{bmatrix} \max_{j \in \mathcal{N}(i)} h_j^{(1)} \\ \max_{j \in \mathcal{N}(i)} h_j^{(2)} \\ \vdots \\ \max_{j \in \mathcal{N}(i)} h_j^{(d)} \end{bmatrix} \quad (3)$$

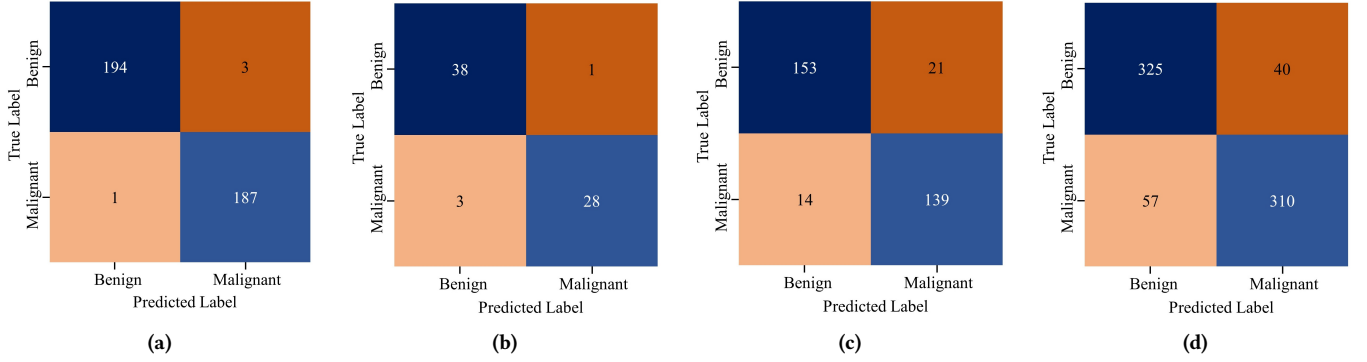
Here,  $h_j^{(d)}$  denotes the  $d^{\text{th}}$  feature dimension of the neighbor node  $j$ . The *mean aggregator* captures the smooth average neighborhood context, encouraging the homogeneity of features between related subspaces, while the *max aggregator* highlights the most discriminatory neighbor activations, promoting feature diversity. In practice, both strategies provide complementary benefits; *mean aggregation* favors stable global reasoning, while *max aggregation* enhances sensitivity to salient subspace relations. By combining self-representation and neighborhood aggregation, the network adaptively fuses subspace-level and contextual information, producing richer, clinically informed embeddings for lesion classification.

**Modeling Long-range Dependencies:** While local aggregation captures the neighborhood-level context between feature subspaces, effectively representing the *global structure* of the lesion requires modeling *long-range dependencies* across distant nodes. To this end, we adopt the Approximate Personalized Propagation of Neural Predictions (APPNP) algorithm [27], which efficiently propagates information through the graph using a personalized PageRank formulation. APPNP approximates topic-sensitive PageRank via iterative propagation, conceptually similar to a *random walk with restarts*, and achieves linear computational complexity with respect to the number of edges. At each power iteration step  $l$ , the node representation is updated as:

$$H^{(l)} = (1 - \alpha) \cdot \tilde{A} H^{(l-1)} + \alpha \cdot H^{(0)} \quad (4)$$

where  $H^{(0)} = \{h_i^K\}$  represents the subspace embeddings obtained after the final layer  $K$  in the previous step in Eqn (2).  $\tilde{A} = \mathcal{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathcal{D}^{-1/2}$  is the normalized adjacency matrix with self-loops, and  $\alpha \in (0, 1)$  is the teleport (or restart) probability that controls the balance between propagation and retention of original information. This iterative scheme allows each node to accumulate information from higher-order neighbors while maintaining sensitivity to its initial subspace representation. The adjacency matrix  $\mathbf{A} \in \mathbb{R}^{N \times N}$  is initialized so that every node is connected to every other node, including itself (self-loop) [28]. Thus, for a fully connected graph, all entries in  $\mathbf{A}$  are set to 1. Inclusion of the self-loop ( $A_{i,i} = 1$ ) ensures that each node preserves its own features during message passing. Before propagation,  $\mathbf{A}$  is row-normalized to stabilize feature aggregation, producing the symmetric normalized adjacency matrix  $\tilde{A}$  and  $\mathcal{D}$  is the degree matrix with  $\mathcal{D}_{i,i} = \sum_j A_{i,j}$ . This normalization balances the influence of neighboring nodes and prevents feature scaling issues during iterative propagation, ensuring stable training even in fully connected graphs.

Importantly, because metadata features are embedded directly within each subspace node, both relation-aware aggregation and



**Figure 2: Confusion matrices illustrating the classification results of the proposed model across four benchmark datasets: (a) ISIC 2024, (b) PAD-UFES-20, (c) HIBA, and (d) HAM10000. Each matrix shows the distribution of true versus predicted labels (benign vs. malignant) using the optimal configuration of graph nodes and metadata features. High values along the diagonal indicate strong classification performance and robust generalization across datasets.**

APPNP propagation jointly capture the interactions between image-derived features (from the CNN backbone) and clinically relevant metadata. This enables the model to learn contextually grounded representations that integrate visual, clinical, and semantic cues.

After the propagation steps  $L$  in Eqn (4), we aggregate the refined subspace embeddings  $H^{(L)}$  by applying the global average pooling across all subspaces (graph nodes). The pooled embedding is then passed through a classifier:

$$\hat{y}_p = \text{Softmax}(\mathbf{W} \cdot \text{AvgPool}(H^{(L)}) + b), \quad (5)$$

where  $\hat{y}_p$  is the predicted class label for image  $p$ , and  $\mathbf{W}$  is a learnable weight matrix, and  $b$  is the bias of the classifier head. The model  $\mathcal{F}$  is trained using a composite objective that combines label-smoothed cross-entropy loss with  $L_2$  regularization to improve stability and reduce overfitting.

## 4 Experiments and Results

**Datasets:** We evaluated our model on four publicly available benchmark skin lesion classifications. **ISIC 2024** [13] contains over 400,000 dermoscopic images extracted from 3D total-body scans, each annotated with detailed lesion- and patient-level metadata. The dataset is highly imbalanced, with 99.9% benign and 0.1% as malignant lesions. **HAM10000** [14] includes 10,015 dermoscopic images (1,824 malignant and 8,191 benign). **PAD-UFES-20** [15] consists of 2,298 smartphone-acquired clinical images (2,246 benign and 52 malignant), representing an extreme class imbalance (2.3% malignant). Finally, **HIBA** [16] consists of 1,635 dermoscopic images (764 malignant and 871 benign).

To mitigate the imbalance, we followed standard SOTA protocols (Table 2): random downsampling of benign cases and augmentation (rotation, flipping, scaling) for malignant samples. Each dataset was divided 80% / 20% into training and testing subsets, maintaining balanced distributions. All experiments were implemented in PyTorch and executed on an AMD Ryzen 7 CPU with 16 GB of RAM. The models were trained for 100 epochs using the Adam optimizer with an initial learning rate of  $1 \times 10^{-3}$ , a batch size of 32, and a weight decay of  $1 \times 10^{-4}$ .

**Table 1: Best achieved skin cancer classification results (%) across the datasets with optimal configurations.**

| Dataset     | Accuracy | Precision | Recall | F1-Score |
|-------------|----------|-----------|--------|----------|
| ISIC2024    | 98.96    | 98.96     | 98.95  | 98.96    |
| PAD-UFES-20 | 94.29    | 94.40     | 94.29  | 94.26    |
| HIBA        | 89.19    | 90.01     | 89.19  | 89.19    |
| HAM10000    | 86.69    | 87.32     | 86.69  | 86.64    |

### 4.1 Results Overview

We evaluated performance using accuracy, precision, recall, and F1-score. The results are reported in three parts: (1) best model performance across datasets, (2) comparison with SOTA approaches, and (3) ablation studies.

Table 1 summarizes the results, and Figure 2 visualizes confusion matrices obtained using optimal metadata and graph configurations. Our model achieved the highest performance on ISIC 2024, with 98.96% accuracy and nearly identical precision, recall, and F1-score. This configuration used six subspace nodes and all metadata features, confirming the synergy between visual and clinical information. The large and diverse ISIC dataset further facilitated robust feature learning and generalization. For PAD-UFES-20, our model achieved an accuracy of 94.29%, with the best configuration using four nodes and all metadata features. The results highlight that a carefully chosen subset of metadata can produce strong performance while mitigating overfitting. Precision slightly exceeded recall, indicating a bias towards minimizing false positives. In HIBA, the model achieved 89.19% accuracy using eight nodes and all metadata, showing that denser graph structures can capture richer intra-sample relations in smaller datasets. Finally, in HAM10000, our approach achieved an accuracy of 86.69% using four nodes and the top 20 metadata features, surpassing previous SOTA methods (Table 2). These results demonstrate that incorporating metadata through graph-based fusion consistently improves classification robustness and accuracy across datasets of varying size and complexity.

**Table 2: Comparison report (%) with SOTA skin lesion classification models across the datasets. Best results are shown in Bold.**

| Dataset      | Model                                       | Accuracy     | Precision    | Recall       | F1-Score     |
|--------------|---------------------------------------------|--------------|--------------|--------------|--------------|
| ISIC Archive | Multi-modal Fusion Model [6]                | 83.80        | —            | —            | —            |
|              | Multi-modal CNN [29]                        | 88.38        | —            | —            | —            |
|              | MetaBlock [10]                              | 91.00        | —            | —            | —            |
|              | Lightweight CNN [5]                         | 92.12        | 92.52        | 91.72        | 91.99        |
|              | CNN-ViT Hybrid [20]                         | 92.50        | 92.07        | 88.85        | 90.38        |
|              | Robust CNN Model [4]                        | 93.48        | 93.24        | 90.70        | 91.82        |
|              | Optimised CNN [30]                          | 94.31        | —            | —            | —            |
|              | Hybrid CNN [31]                             | 94.41        | 94.00        | 95.00        | 94.00        |
|              | Multi-modal Network [32]                    | 94.90        | —            | —            | —            |
|              | CNN-Attention Hybrid [19]                   | 96.70        | 96.69        | 97.61        | 96.69        |
|              | <b>Proposed</b>                             | <b>98.96</b> | <b>98.96</b> | <b>98.95</b> | <b>98.96</b> |
| PAD-UFES-20  | MetaBlock [10]                              | 77.00        | —            | —            | —            |
|              | Multi-modal Fusion Model [6]                | 77.20        | —            | —            | —            |
|              | MDFNet [33]                                 | 80.42        | —            | —            | —            |
|              | Self-contrastive Multimodal [34]            | 83.30        | 83.20        | 83.50        | 83.00        |
|              | Multi-modal Network [32]                    | 84.80        | —            | —            | —            |
|              | Optimised CNN Model [35]                    | 84.90        | —            | —            | —            |
|              | Multi-modal Contrastive [23]                | 89.60        | —            | —            | 90.00        |
|              | CNN-Attention Hybrid [19]                   | 91.02        | 91.35        | 91.05        | 91.19        |
|              | <b>Proposed</b>                             | <b>94.29</b> | <b>94.40</b> | <b>94.29</b> | <b>94.26</b> |
| HIBA         | Transformer-Based Deep Neural Networks [36] | 84.71        | —            | —            | —            |
|              | CNN Model [37]                              | 87.00        | 86.00        | 85.40        | 85.70        |
|              | Multi-modal Contrastive [23]                | 87.50        | —            | —            | 88.60        |
|              | <b>Proposed</b>                             | <b>89.19</b> | <b>90.01</b> | <b>89.19</b> | <b>89.19</b> |
| HAM10000     | Multi-modal Transformer [38]                | 81.60        | —            | —            | —            |
|              | Multi-modal Neural Network [39]             | 85.20        | 85.20        | —            | 85.20        |
|              | Multi-modal Contrastive [23]                | 85.90        | —            | —            | 61.80        |
|              | <b>Proposed</b>                             | <b>86.69</b> | <b>87.32</b> | <b>86.69</b> | <b>86.64</b> |

Dashed entries (—) denote metrics not reported in the original publication.

## 4.2 Comparison With State-of-the-art (SOTA)

Table 2 summarizes the comparisons against strong baselines in all four datasets. Overall, our graph-driven multimodal model consistently outperforms prior methods, with gains attributable to (i) early, subspace-level metadata fusion into the same latent space as visual features, (ii) inductive neighbourhood aggregation that corrects ambiguous visual evidence, and (iii) global propagation that captures long-range dependencies. Compared to [19] (HierAttn), the accuracy improves from 96.70% to 98.96% on ISIC (+2.26) and from 91.02% to 94.29% on PAD-UFES-20 (+3.27). HierAttn’s deeply supervised multiscale attention enhances image features but treats metadata peripherally. In contrast, our subspace-level fusion allows clinical attributes to actively reshape representations during message passing, improving both precision and recall.

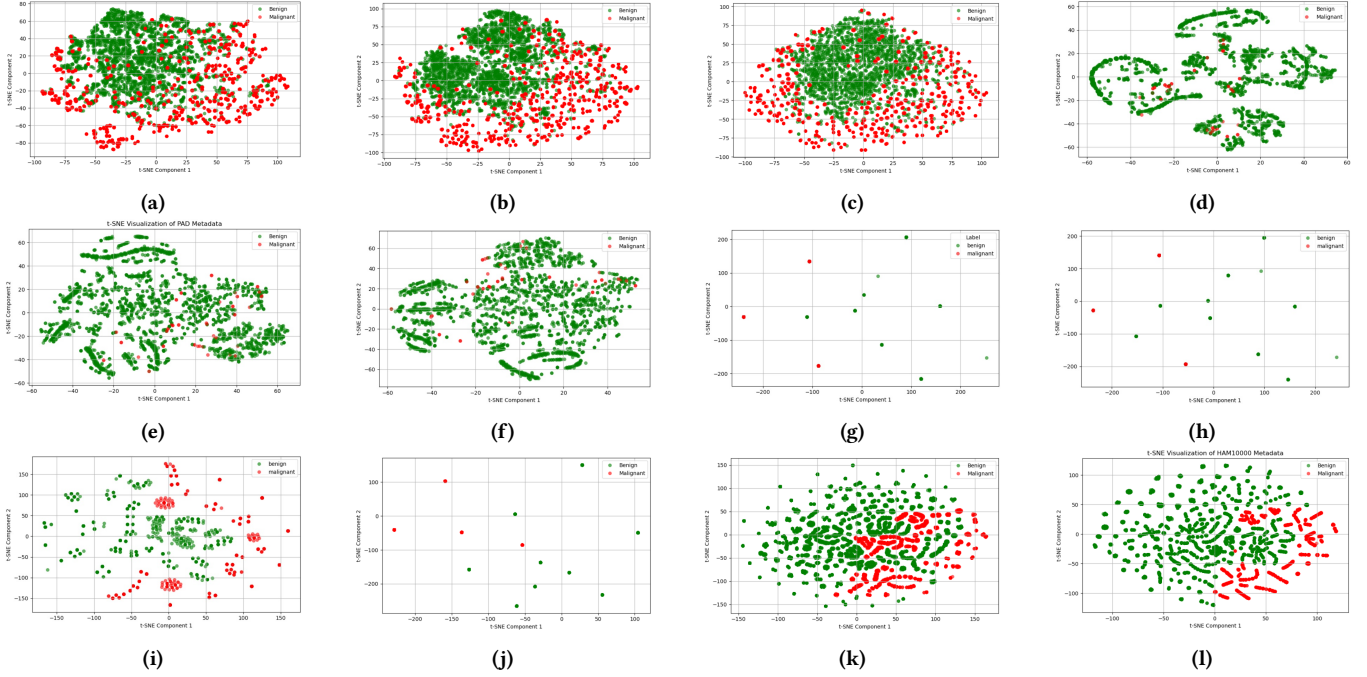
Compared with the nested multimodal contrastive pretraining of [23] (SLIMP), we observe larger gains where relational fusion is critical: on PAD-UFES-20, accuracy rises from 89.60% to 94.29% (+4.69); on HIBA, from 87.50% to 89.19% (+1.69); and on HAM10000, from 85.90% to 86.69% (+0.79). SLIMP primarily strengthens representations through pretraining, while our approach continues to integrate metadata relationally during training, yielding additional context-aware gains.

On HIBA, our model surpasses transformer- or scale-oriented and CNN baselines, including [36] (84.71% → 89.19%, +4.48) and [37] (87.00% → 89.19%, +2.19). These results suggest that explicit intra-embedding relational modeling offers advantages beyond large-scale self-attention when metadata and inter-feature relationships matter.

Across ISIC baselines, MetaBlock [15], lightweight CNNs [5], CNN-ViT hybrids [20], and optimized/hybrid models [32], our accuracy improvements range from roughly +4.1 to +8.0 percentage points. Taken together, these findings indicate that combining (i) metadata embedding in the shared latent space, (ii) subspace partitioning with inductive neighbourhood aggregation, and (iii) global propagation via APPNP systematically improves class separability and robustness to dataset shift, especially for visually ambiguous or under-represented classes.

To ensure a fair and reliable comparison, all reported SOTA baseline results are reported in the respective papers. These results are based on the evaluation protocols, such as standard data splits (e.g., 80%/20%), which align with our own evaluation setup. However, the key distinction lies in the training methodology. Most cited baselines rely on extensive fine-tuning of large backbones such as ResNet or EfficientNet, or on resource-intensive pre-training, as in SLIMP [23].





**Figure 3: Visualizations of feature embeddings using t-SNE using different numbers of metadata features: (a) ISIC with top 10 features, (b) ISIC with top 20 features (c) ISIC with top 40 features, (d) PAD with top 10 features, (e) PAD with top 25 features (f) PAD with top 30 features, (g) HIBA with top 10 features (h) HIBA with top 20 features, (i) HIBA with top 25 features, (j) HAM with top 15 features, (k) HAM with top 20 features, and (l) HAM with top 30 features. These plots illustrate that increasing the number of metadata features enhances the separability between benign (green) and malignant (red) samples, indicating that incorporating a richer set of metadata features contributes to improved discriminability.**

Our approach uses a frozen pre-trained CNN backbone. This design extracts embeddings once, and all trainable parameters are restricted exclusively to the GNN module. This makes the model substantially more computationally efficient [5]. Consequently, the superior performance of our proposed approach highlights the effectiveness of frozen-backbone adaptation and graph-driven fusion compared with costly end-to-end fine-tuning.

**Table 3: Accuracy (%) across varying subspaces (nodes) with and without metadata. Best results are shown in Bold**

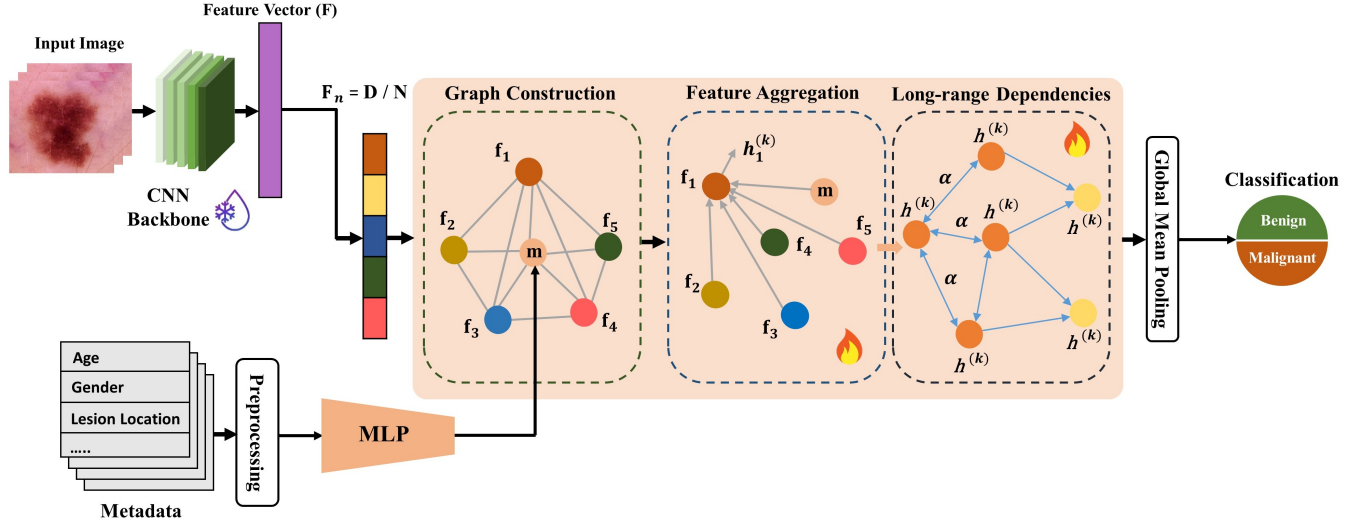
| Metadata Nodes # |    | ISIC2024     | PAD-UFES-20  | HIBA         | HAM10000     |
|------------------|----|--------------|--------------|--------------|--------------|
| No Metadata      | 4  | 98.44        | 84.29        | 75.00        | 79.18        |
|                  | 6  | 98.18        | 82.14        | 79.05        | 78.50        |
|                  | 8  | 98.45        | 71.43        | 77.70        | 78.84        |
|                  | 16 | 95.30        | 68.25        | 74.32        | 67.27        |
|                  | 32 | 94.70        | 63.49        | 66.89        | 73.33        |
| With Metadata    | 4  | 98.44        | <b>94.29</b> | 85.81        | 81.42        |
|                  | 6  | <b>98.96</b> | 83.93        | 83.78        | 86.01        |
|                  | 8  | 98.70        | 85.71        | <b>89.19</b> | <b>86.69</b> |
|                  | 16 | 97.85        | 69.84        | 77.70        | 74.55        |
|                  | 32 | 96.88        | 58.73        | 76.35        | 74.24        |

### 4.3 Ablation Study

We conducted extensive ablations to assess the influence of key design choices: number of subspaces (nodes), metadata selection, architectural components, and late fusion variants.

**Effect of metadata and node count:** Table 3 shows that incorporating metadata consistently improves performance across all datasets. For example, the PAD-UFES-20 accuracy improved from 84.29% to 92.86% when metadata were integrated using four subspaces. Similarly, ISIC 2024 reached its peak (98.96%) using metadata and six nodes. However, beyond a moderate node count (e.g., 16–32), performance degraded and probably due to overfitting from excessive relational updates or spurious feature correlations.

**Metadata feature selection:** Table 4 further explores how varying the number of metadata features affects performance. We selected the main metadata features using the random forest feature importance technique [40] and visualized them using t-SNE [41] as shown in Figure 3. The results show that while ISIC 2024 consistently benefits from using all metadata, other datasets, such as HAM10000, achieved their best accuracy of 86.69% using only the top 20 metadata features with four subspaces. This indicates that the selective use of metadata is more effective in smaller or noisier datasets, where irrelevant metadata may introduce noise rather than improve discrimination. Furthermore, PAD performs best with all metadata and fewer nodes, while HIBA achieves a top accuracy



**Figure 4: Alternative configuration in which the metadata-derived MLP output ( $m$ ) is treated as a dedicated subspace node within the graph. The whole process is the same as in Figure 1, while the MLP processes metadata to produce a compact embedding that forms an additional node ( $N+1$ -th node). This metadata node is connected to all other image nodes in a fully connected graph, enabling message passing between visual and clinical representations during graph learning.**

of 89.19% using all metadata with eight subspaces. These variations suggest that both metadata richness and graph density should be tailored to the characteristics of each dataset.

**Table 4: Accuracy (%) for the best metadata feature selection on four skin lesion datasets. Best results are shown in Bold.**

| Dataset     | Top 10 | Top 20       | Top 30 | All          |
|-------------|--------|--------------|--------|--------------|
| ISIC 2024   | 98.43  | 98.69        | 98.70  | <b>98.96</b> |
| PAD-UFES-20 | 91.43  | 92.86        | 90.00  | <b>94.29</b> |
| HIBA        | 79.27  | 81.71        | 79.87  | <b>89.19</b> |
| HAM1000     | 81.15  | <b>86.69</b> | —      | 80.87        |

**Component analysis:** Table 5 examines the impact of different architectural choices. The use of Min aggregation consistently outperformed Max aggregation, and the disabling of additional projection layers often led to better results. For example, the configuration using only the propagation layer achieves competitive or superior performance in most datasets, including 81.10% for HIBA and 80.87% for HAM10000. These results suggest that simpler graph designs, relying primarily on relational message passing, are often more effective than deeper or more complex architectures, especially when combined with well-structured metadata. These ablation studies demonstrate that the integration of metadata, thoughtful subspace (node) selection, and lightweight architecture plays a crucial role in achieving robust and generalizable performance. They highlight the need for dataset-specific tuning, as the optimal configuration varies significantly between datasets with different sizes, class distributions, and metadata quality.

**Late fusion with subspace variant:** We further experimented with a *late-fusion variant* of our model to compare against the proposed early graph fusion. In this design, metadata were first passed

**Table 5: Accuracy (%) of various architectural components across the datasets. Best results are shown in Bold.**

| Component              | ISIC2024     | PAD-UFES-20  | HIBA         | HAM10000     |
|------------------------|--------------|--------------|--------------|--------------|
| Min Aggregation        | 97.92        | 85.71        | 74.39        | 78.41        |
| Max Aggregation        | 95.31        | <b>88.57</b> | 72.56        | 76.50        |
| Layer Project (True)   | 98.20        | 81.43        | 71.95        | 74.32        |
| Only Propagation Layer | <b>98.43</b> | 87.14        | <b>81.10</b> | <b>80.87</b> |

**Table 6: Accuracy (%) using MLP feature extraction for metadata and the late fusion technique with varying numbers of nodes. Best results are shown in Bold.**

| Node # | ISIC2024     | PAD-UFES-20  | HIBA         | HAM10000     |
|--------|--------------|--------------|--------------|--------------|
| 2      | 98.21        | 70.00        | 83.54        | 82.79        |
| 4      | 97.81        | 71.43        | 84.51        | <b>84.15</b> |
| 8      | <b>98.68</b> | 71.43        | 84.76        | 76.23        |
| 16     | 97.37        | <b>72.86</b> | <b>85.98</b> | 74.59        |
| 32     | 93.86        | 65.71        | <b>85.98</b> | 68.30        |

through an MLP to learn a high-level embedding (dotted path in Fig. 1) and concatenated with graph-derived image features from the frozen CNN backbone just before the classification layer. This decouples metadata processing from graph propagation, allowing independent representation learning. As shown in Table 6, this approach produced competitive results in all datasets. For example, the ISIC 2024 dataset achieved 98.68% accuracy with eight nodes, which is comparable to the best early fusion result (98.96%). Similarly, HIBA achieved an accuracy of 85.98% using 16 and 32 nodes, suggesting that metadata embeddings from the MLP contributed positively to classification, especially in smaller datasets where metadata could provide stronger discriminative power. However,



PAD-UFES-20 and HAM10000 showed mixed performance, with accuracies peaking at 72.86% and 84.15%, respectively, and degrading at higher node counts. This is due to the loss of relational structure when metadata is fused too late in the pipeline, especially if its influence is diminished relative to the graph signal.

Overall, while MLP-based late fusion provides a valid alternative, particularly for semantically rich metadata, graph-driven early fusion consistently yields higher accuracy and stronger generalization. This confirms that propagating clinical information through graph message passing is more effective for robust and interpretable lesion classification. However, late fusion could still be beneficial in specific scenarios, particularly when metadata is highly structured and semantically rich, and its independent representation learning can add complementary information not captured through joint embedding.

**Alternative graph-level metadata integration:** We further explored an alternative design in which the *metadata-derived MLP* [42] *output* was incorporated directly into the graph as an *additional subspace node*, rather than being concatenated at the feature level (Figure 4). In this configuration, the MLP produced a compact embedding of the metadata, which was appended as a new node connected to all existing subspace nodes derived from CNN features. This design allowed *direct message passing between visual and metadata nodes*, facilitating potential cross-modal interactions at the graph level rather than through simple vector concatenation. However, empirical results (Table 7) revealed that this configuration led to a notable decline in classification accuracy compared to the baseline early-fusion model, where metadata are concatenated with CNN features before graph partitioning. We attribute this degradation to a *representation-scale imbalance*: CNN-derived subspaces encode dense, high-dimensional visual information, while the metadata node is comparatively low-dimensional and semantically distinct. This discrepancy caused the GNN to underutilize the metadata node during message aggregation, thereby disturbing the learned inter-subspace relations and reducing the overall discriminative power.

**Table 7: Comparison (%) of early and late fusion (Figure 1) with the processed metadata (MLP features) as a separate node, as shown in Figure 4. Best results are shown in Bold.**

| Dataset     | MLP as Node | Early Fusion | Late Fusion |
|-------------|-------------|--------------|-------------|
| ISIC2024    | 98.25       | <b>98.96</b> | 98.68       |
| PAD-UFES-20 | 62.86       | <b>94.29</b> | 72.86       |
| HIBA        | 78.66       | <b>89.19</b> | 85.98       |
| HAM10000    | 80.87       | <b>86.69</b> | 84.15       |

Table 8 presents the best classification accuracy achieved by four frozen CNN architectures: ResNet152, EfficientNet, VGG 19, and InceptionNet, across four benchmark skin lesion datasets with metadata integration. For each model, the highest accuracy obtained across different metadata node configurations was selected. ResNet152 consistently outperformed other models, achieving the highest accuracy on all datasets, including 98.96% effectiveness of metadata fusion and the strong performance of deeper architectures such as ResNet152 in skin lesion classification tasks.

**Table 8: Best accuracy (%) for each CNN model across datasets with metadata integration. Best results are shown in Bold.**

| Dataset     | ResNet152    | EfficientNet | VGG 19 | InceptionNet |
|-------------|--------------|--------------|--------|--------------|
| ISIC2024    | <b>98.96</b> | 95.31        | 94.27  | 92.97        |
| PAD-UFES-20 | <b>94.29</b> | 91.16        | 91.07  | 91.09        |
| HIBA        | <b>89.19</b> | 79.88        | 77.44  | 81.71        |
| HAM10000    | <b>86.69</b> | 83.61        | 77.32  | 76.23        |

## 5 Conclusion

We proposed a *graph-driven multimodal framework* for skin lesion classification that seamlessly integrates deep visual representations with structured clinical metadata through *subspace-level fusion* using graph neural networks (GNNs). By employing a *frozen CNN backbone*, the model eliminates the need for costly fine-tuning while achieving robust adaptation to the dermatological domain through graph-based reasoning over intra-image and inter-modal relationships. The framework constructs *fully connected graphs* on partitioned CNN feature subspaces, enriched with patient metadata, allowing *message passing and contextual refinement* within and between modalities. Extensive experiments on four benchmark datasets: ISIC 2024, HAM10000, PAD-UFES-20, and HIBA, demonstrating consistent gains over state-of-the-art approaches in accuracy, precision, recall, and F1-score.

Ablation studies further validate the effectiveness of metadata-driven fusion, showing that combining visual and semantic cues yields more discriminative and generalizable representations. In general, this work underscores the potential of graph-based multimodal fusion as a principled and scalable approach to clinically interpretable medical image analysis, paving the way for more trustworthy and context-aware AI-assisted diagnostic systems.

We plan to extend this framework to multi-view, temporal dermatological, and real-world clinical data, exploring dynamic graph construction for evolving clinical contexts and cross-institutional generalization to further enhance the robustness and clinical reliability of graph-based medical AI models.

## Acknowledgments

This work was supported by UK Research and Innovation (UKRI) – Economic and Social Research Council (ESRC) under the SCAnDi project (Grant: ES/Y010655/1) and the Research Investment Fund (RIF) from Edge Hill University (Grant: 1REWAB25).

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